HAOMING WANG

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RESEARCH INTERESTS

Multimodal Foundation Models, Vision-Language Models, Visual Spatial Reasoning, Instruction-Conditioned World Generation, RL Post-Training, Embodied AI Decision-Making.

EDUCATION

Sept 2022 – May 2027 (Anticipated) **Ph.D. in Electrical and Computer Engineering**
University of Pittsburgh, Pittsburgh, PA, USA
Advisor: Prof. Wei Gao, Intelligent System Lab

Sept 2018 – May 2022 **B.Eng. in Automation** GPA: 3.8/4.0
Zhejiang University, Hangzhou, China
Chu Kochen Honors College

EXPERIENCE

Sept 2022 – Present **Graduate Student Researcher**
Intelligent System Lab, University of Pittsburgh, Pittsburgh, PA, USA

- **(May 2025 – Present)** Led research on multimodal foundation models: an LLM-agent + procedural 3D world generator with verifiable reward (CVPR'26 Oral); Dirichlet-energy latent shaping of VLM 3D-topology subspaces; sequential observation-budget policies for cross-frame video reasoning.
- **(Nov 2024 – May 2025)** Drove generative-model interpretability and on-device LLM personalization: training-free fine-grained explanation of diffusion / GAN / autoregressive image generators (FineXL); explainable model selection for warm-started LLM personalization (MobiSys'25 Xpert).
- **(Sept 2022 – Oct 2024)** Designed federated learning systems for heterogeneous AIoT fleets: gradient correction under data / device heterogeneity (MobiCom'25); FL with unbounded device staleness (AAAI'25).

PUBLICATIONS (PEER-REVIEWED)

1. **[CVPR'26 Oral] InfiniBench: Infinite Benchmarking for Visual Spatial Reasoning with Customizable Scene Complexity**
   Haoming Wang, Qiyao Xue, Wei Gao.
   IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026 (Acceptance 25.4%, Oral).
   [arXiv] [Code] [Data]
   - **Architected** a customizable framework for analyzing VLM spatial reasoning that generates an unlimited supply of photo-realistic 3D test scenes; the LLM-agent + procedural-constraint feedback loop is a clean recipe for wrapping a language model around a constrained generator with verifiable reward – directly relevant to instruction-conditioned world generation.
2. **[MobiSys'25] Never Start from Scratch: Expediting On-Device LLM Personalization via Explainable Model Selection**
   Haoming Wang, Boyuan Yang, Xiangyu Yin, Wei Gao.
   Proceedings of the 23rd ACM International Conference on Mobile Systems, Applications and Services (MobiSys), 2025 (Acceptance 18.0%). [Paper]
   - **Developed Xpert**, an explainable model-selection method that warm-starts on-device LLM personalization from the closest pre-personalized checkpoint of a similar past user.

3. **[MobiCom'25] When Device Delays Meet Data Heterogeneity in Federated AIoT Applications**
   Haoming Wang, Wei Gao.
   Proceedings of the 31st ACM International Conference on Mobile Computing and Networking (MobiCom), 2025 (Acceptance 17.1%). [Paper]
   - **Engineered** a federated AIoT training method that jointly disentangles and corrects for data heterogeneity and device delays via targeted gradient correction.
4. **[AAAI'25] Tackling Intertwined Data and Device Heterogeneities in Federated Learning with Unlimited Staleness**
   Haoming Wang, Wei Gao.
   Proceedings of the 39th AAAI Conference on Artificial Intelligence, 2025 (Acceptance 23.4%). [arXiv]
   - **Designed** a gradient-inversion technique that recovers usable training signal from arbitrarily stale device updates, enabling FL to converge under unbounded staleness.

Preprints & Papers Under Review

1. **Uncovering and Shaping the Latent Representation of 3D Scene Topology in Vision-Language Models**
   Haoming Wang, Wei Gao.
   Preprint, 2026. [Paper] [Code]
   - **Discovered** a VLM latent subspace encoding 3D scene topology and **introduced** a Dirichlet-energy regularizer that improves real-world spatial reasoning by up to 12.1% over standard SFT – a representation-level signal complementary to RL post-training.
2. **MosaicThinker: On-Device Visual Spatial Reasoning for Embodied AI via Iterative Construction of Space Representation**
   Haoming Wang, Qiyao Xue, Weichen Liu, Wei Gao.
   Under Review, 2026. [arXiv]
   - **Built** a training-free pipeline that stitches multi-view RGB frames into a compact semantic map; the iterative key-frame-selection step is a sequential information-acquisition policy under an observation budget – sister problem to chunked / autoregressive generation. Up to 40% accuracy gain.
3. **Reasoning Path and Latent State Analysis for Multi-view Visual Spatial Reasoning: A Cognitive Science Perspective**
   Qiyao Xue, **Haoming Wang** (co-author), Weichen Liu, Shiqi Wang, Yuyang Wu, Wei Gao.
   Under Review, 2026. [arXiv]
   - **Co-authored** ReMindView-Bench (50K+ VQAs across 15 VLMs); linear-probe + entropy-dynamics analysis yields an uncertainty signal that separates correct vs. incorrect reasoning trajectories – a candidate reward signal for RL post-training.
4. **HEVEN: Small Object Search and Navigation on VLM-Based Autonomous Mobile Systems**
   Anonymous Authors (incl. **Haoming Wang**, 3rd author).
   Double-Blind Under Review, 2026.
   - **Co-architected** the first VLM-based mobile system for small-object search, with a hierarchical spatial semantic tree and Tree-of-Thought planner + asynchronous speculative execution; a concrete instance of long-horizon multimodal decision-making under anytime constraints.
5. **GlobalNav: Daily Object Navigation in VLM-Based Autonomous Mobile Systems with Aligned Local and Global Views**
   Anonymous Authors (incl. **Haoming Wang**, 3rd author).
   Double-Blind Under Review, 2026.
   - **Co-developed** an infrastructure-assisted VLM navigation system aligning ceiling-camera global views with the robot's first-person view; lifted daily-object navigation success from <30% to 90.91% – a natural example of RL-style replanning under partial observation.
6. **Deciphering Personalization: Towards Fine-Grained Explainability in Natural Language for Personalized Image Generation Models (FineXL)**
   Haoming Wang, Wei Gao.

Under Review, 2025. [Paper]

- **Architected FineXL**, a training-free method decomposing a personalized image generator's divergence from its base into orthogonal natural-language concepts; cut explanation error by 56% across diffusion, GAN, and autoregressive generators.

7. **FreezeAsGuard: Mitigating Illegal Adaptation of Diffusion Models via Selective Tensor Freezing**
   Kai Huang, **Haoming Wang** (co-author), Wei Gao.
   Preprint, 2024. [Paper] [Code]
   - **Co-developed** a bilevel-optimization defense that learns a tensor-freezing mask over Stable Diffusion – a differentiable-mask recipe applied to diffusion training; 37–47% stronger mitigation than unlearning baselines.
8. **Spatial Reasoning in Multimodal Large Language Models: A Survey of Tasks, Benchmarks and Methods**
   Weichen Liu, Qiyao Xue, **Haoming Wang**, Xiangyu Yin, Boyuan Yang, Wei Gao.
   Under Review, 2026.

Honors & Awards

April 2026 — **ECE Research Assistant of the Year**
Department of Electrical and Computer Engineering, University of Pittsburgh

Technical Skills

Models & Methods: Stable Diffusion, CLIP, Gemini, GPT-4o/5, InternVL, Qwen-VL, LLaVA-Video; bilevel optimization, Tree-of-Thought planning, sequential information-acquisition policies.

Simulation & 3D: Habitat, Isaac Sim, Infinigen, Blender; frontier-based camera planning, SAM-3D / 3DGS, DepthPro.

Programming: Python, PyTorch, CUDA, C/C++.